Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/24/2025
PI Name:	Wu, Shu-Chieh		
Project Title:	Human Systems Integration Architecture: Concept Validation and Implementation Evaluation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	95192	**Congressional District:**	17
Comments:			
Project Type:		**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	10/01/2022	**End Date:**	10/01/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Shu-chieh (San Jose State University Research Foundation Inc) McTigue, Kaitlin M.S. (NASA Ames Research Center) Vera, Alonso Ph.D. (NASA Ames Research Center) Parisi, Megan (NASA Ames Research Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The likelihood of an unanticipated major vehicle malfunction during all Design Reference Missions (DRMs) is estimated to be greater than 10% for a 30-day mission, based on analysis of extended low-Earth Orbit (LEO) and Lunar human spaceflight mission data. The consequences of such unanticipated major vehicle malfunctions become more significant with increasing Distance from Earth, as the primary countermeasures currently available rely on support from experts at mission control on the ground (e.g., verbal guidance, troubleshooting) delivered through real-time communication. The possibility of adverse performance outcomes given decreasing real-time ground support during future exploration missions, most consequentially that flight crew are unable to adequately respond to unanticipated major vehicle malfunctions and execute safety-critical procedures, is characterized as a Human Systems Integration Architecture (HSIA) risk. Successful mitigation will require seamless collaboration between flight crew and onboard vehicle capabilities to enable effective and efficient responses to the aforementioned situations. Identification of onboard vehicle capabilities and crew roles and responsibilities necessary to achieving effective human-systems collaboration will require iterative cycles of concept development and empirical evaluation of human performance in complex operations. This task addresses Gaps HSIA-401 and HSIA-501 (and alternatively Gaps HSIA-101, HSIA-201, and HSIA-701) but also has general implications for other HSIA Gaps by informing the evaluations of potential countermeasures.
Rationale for HRP Directed Research:	This proposal qualifies for a directed study due to the urgent, time-sensitive need to develop Human Systems Integration Architecture (HSIA) risk characterization research for Artemis missions and beyond. Timely data analysis in suitable simulations can inform future Artemis missions, as well as missions to Mars. In order to meet the NASA Human Research Program (HRP) milestones for lunar and Mars transit missions, and to establish a baseline human performance threshold, this work must be directed due to insufficient time for solicitation. Additionally, the directed nature of this study allows the HSIA Discipline Scientist to leverage previously collected data that may inform solutions.
Research Impact/Earth Benefits:	Our research program and associated deliverables will contribute to reducing the Earth Independent Human-Systems Operations (EIHSO) risk, formerly known as Human-Systems Integration Architecture (HSIA) risk. They will also inform gaps in knowledge relevant to maintaining crew health and performance on future long duration exploration missions beyond low-Earth orbit (LEO).
Task Progress:	Final report was submitted to the Human Factors and Behavioral Performance (HFBP) Element in January 2024. The final report was edited and released as NASA TM (NASA/TM–20240011157) in August 2024. Primary results: A survey conducted as part of the task found that most NASA simulation facilities are designed to achieve high physical fidelity while Human Systems Integration Architecture (HSIA) risk mitigation requires simulation emphasizing task and functional fidelity aspects. A trade analysis shows that, for standard and requirement development, evaluation conducted using synthetic task environments with inexperienced participants will support testing a wide variety of conditions and yield findings robust enough to be generalized to a wide variety of designs on which developed standards and requirements might be levied, while allowing human performance standard measures to be collected using consistent methods across tasks and conditions. For technology/tool development, because findings will only need to be generalized to the actual target environment in which the technology/tool will be used, it is more suitable to evaluate the prototypes in a scaled world that preserves functional relationships present in the actual target environment with intended user populations.
Bibliography Type:	Description: (Last Updated: 06/24/2025)
NASA Technical Documents	Wu S-C, Parisi ME, McTigue KR, Robertson IW, Panontin TL, Vera, AH. "Determining simulation fidelity necessary for evaluating onboard vehicle capabilities and crew roles on long duration exploration missions beyond low-Earth orbit." Moffett Field, California: Ames Research Center, 2024. NASA/TM–20240011157. https://ntrs.nasa.gov/citations/20240011157 , Aug-2024